

P101 | Unit 2 | Quick reference

Measurement, variation and justified precision

Recall fluently

Know the distinctions well enough to explain them in your own words. A digit or a summary is meaningful only with its quantity, unit and method.

Quantity and procedure	Define the measurand: the quantity intended for measurement, including endpoints and object state. Distinguish an endpoint separation from a distance along a curve.
Fresh repeats	Repeat the stated procedure, such as lifting and replacing a ruler. Re-reading an unchanged placement or copying an entry provides different evidence. Keep chronological order.
Resolution and recording	Resolution concerns distinguishable changes in indication. State instrument divisions and the chosen recording increment. Extra printed digits do not establish finer resolution or better accuracy.
Repeatability and accuracy	A tightly grouped set describes agreement under the repeat conditions. It can still share a systematic discrepancy. Accuracy requires a relationship to the quantity being measured; a suitable reference helps investigate it.
Variation and shared effects	Random variation changes among repetitions; a systematic effect can persist or vary predictably. A small observed spread does not show that systematic effects are absent.
Missing is not zero	An absent reading supplies no number. A numerical zero may be valid. A documented transcription correction needs source evidence; an unusual value alone is not proof of invalidity.
Preserve provenance	Keep source, first transcription, decisions with reasons, processed values and report linked by IDs. Excluding a value from one calculation does not delete its source record.

Readings are evidence only when their quantity, method and history remain visible.

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Understand and reconstruct

For n included finite real readings of the same quantity in one common unit, with n greater than zero. Every observation counts, including repeated numerical values.

$$\bar{x} = \frac{x_1 + \dots + x_n}{n}$$

$$R = x_{\max} - x_{\min}$$

$$D = \frac{|x_1 - \bar{x}| + \dots + |x_n - \bar{x}|}{n}$$

Arithmetic mean	Add all included values and divide by their count. The mean need not equal an observed value. Its extra arithmetic digits do not establish a new instrument capability.
Range	Subtract the minimum from the maximum. The observed interval gives both endpoints; the range is their difference. It does not describe bounds on future readings.
Mean absolute deviation	D is the mean of the absolute distances from the arithmetic mean. Use that same mean for every distance. Adding signed deviations instead gives cancellation, not a spread.
Worked check	For 2, 4, 6 mm: mean 4 mm; range 4 mm; distances 2, 0, 2 mm; D is four-thirds of a millimetre (about 1.33 mm). All three summaries carry the readings' unit.
Conditions and limits	An empty list has no mean, range or D. A singleton has range and D zero but cannot reveal repeatability. These descriptive spreads are not a guaranteed error bound, confidence interval or complete measurement uncertainty.
Shifts and unit changes	Adding the same constant to all values shifts the mean equally but leaves range and D unchanged. Converting mm to m divides values, mean, range and D by 1000. Two endpoint coordinates cancel the same additive offset in their difference; a direct length indication does not.

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Consult while working

Use this page as a practical aid. You need to understand the decisions, but do not need to memorise Python syntax or a reporting template.

A comparison	State the question, changed feature and controls. Preserve order, inspect each method separately and use common graph scales. A different mean is not automatically better accuracy or proof of a cause.
An audit trail	Count attempts separately from included readings. Preserve missing entries; record every correction or exclusion and its evidence. Retain unusual values when no independent invalidity reason is supplied.
Rounding	Keep working digits and round once to the final display. This unit uses nearest with exact ties to even: 7.245 to two decimal places is 7.24; 7.255 becomes 7.26. Binary floats may not represent decimal ties exactly.
Digits and units	Decimal places count positions after the point. Significant digits begin at the first nonzero digit and include intended trailing digits. 0.00470 m and 4.70 mm both express three significant digits. Exact counts and defined conversions do not supply measurement uncertainty.
Python record handling	Use dictionaries with stable IDs, units and notes. None represents missing; test value is None, not whether value is zero. Build a new processed list from explicit decisions. The lesson functions assume finite real inputs in a common unit.
Python summaries and display	Use the notebook mean_value, range_value and mean_absolute_deviation functions after eligibility decisions. Refuse an empty numerical summary. format(value, ".1f") returns display text; keep the numerical working value.
A defensible report	Identify source/simulated status, measurand, method, units, attempt and inclusion counts, decisions, labelled summaries and limitations. Propose a check tied to a stated limitation. Supplied-data analysis does not demonstrate apparatus handling.

Use alongside Lessons 1-5, the investigation notebook and Unit 2 exercises. The reference is a lookup aid, not a substitute for the worked reasoning.

librauni.github.io/learn/physics/stage-1/p101/b01/u02/